Response to Reviewers

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Opening Remarks

The authors sincerely thank the Editorial Board and the reviewers for the time and effort devoted to handling and reviewing our manuscript, as well as for their constructive and insightful comments. All comments and concerns raised by the reviewers have been carefully addressed in a point-by-point manner. For ease of reference, the following color-coded convention is used throughout this document:

- **Editors' and Reviewers' original comments** are displayed in dark blue.
- **Authors' responses** are preceded by the green term "Response."
- **Modified text in the manuscript** is presented as block quotations in maroon.

In addition, we have included a highlighted copy of the revised manuscript in which all modifications are clearly indicated. We hope that the revised version of the manuscript adequately addresses the concerns raised by the reviewers and meets the high standards of the *IEEE/ASME Transactions on Mechatronics*.

Response to Senior Editor

Comment: I agree with the TE's assessment.

Response: Thank you for your comment.

Response to Technical Editor

Comment: This paper presents a control method for aerial continuum manipulators with physical interaction. Three expert reviewers have assessed the manuscript. The reviewers have identified several critical problems that prevent acceptance of the paper. For example:

- Limited conceptual novelty at the control-architecture level and insufficient reformulation of hybrid vision/force methods to address the strong vehicle–arm coupling, configuration-dependent compliance and comparable time-scales of the TD-ACM dynamics.

- Strong modelling assumptions (full-rank Jacobians, negligible friction, simplified spring-like contact, fixed plane parameters) that are neither rigorously justified nor verified in simulation or experiment, together with inconsistencies in the interaction-matrix inversion.

- Incomplete experimental evaluation: feasibility is shown, but no hardware comparison against the PD-PI or CISMC baselines used in simulation; critical controller parameters, latency and failure cases of the feature extractor are also missing.

- Inadequate treatment of feature appearance/disappearance, transitions between free-motion and contact, and the consequences of temporary rank deficiency when fewer than three lines are available.

The Editorial Board agrees with these comments. It is also noted by the Board that the submitted video shows experimental video, however, these are very simple results compared to state-of-the-art methods; this simplicity doesn't help to make a strong case for the proposed work. For these reasons, this paper can't be considered in the present form, as it requires extensive revisions. If the authors decide to resubmit their work, they need to address all these issues in a point-by-point manner.

Response: The authors appreciate the Technical Editor's insightful comments. In the revised manuscript, the novelty of the proposed control architecture has been more clearly highlighted. In particular, the adaptive super-twisting control law, integrated with the proposed projection-based fixed-time sliding-mode controller, provides enhanced robustness against uncertainties and disturbances. Furthermore, the manuscript introduces novel coupled dynamic equations for the TD-ACM, for which additional derivation and discussion have been provided in the revised version. These equations explicitly capture the vehicle–arm dynamic coupling and configuration-dependent compliance, thereby providing the foundation for the proposed compliance-control framework.

Several strong modeling assumptions have also been relaxed. In particular, friction effects have now been incorporated into the dynamic equations, while image-space singularities are addressed through the integration of a singularity-aware line selection algorithm (SALS). The assumption of fixed plane parameters has also been further justified by introducing a procedure in which the interaction matrix is evaluated at the desired camera configuration. Consequently, during operation, deviations from the desired configuration are effectively captured as uncertainty in the interaction matrix. In addition, the inversion of the interaction matrix is now treated consistently throughout the manuscript using the Moore–Penrose pseudoinverse.

Furthermore, the hardware experiments have been repeated for the PD-PI and CISMC controllers, and their performance is compared using the RMSE and standard deviation (STD) metrics. The critical controller parameters have also been added to the manuscript. To robustly handle transitions between free-motion and contact modes, a hysteresis-based contact detection algorithm (CDA) and a robust adaptive Kalman filter (RAKF) have been integrated into the control framework. Finally, two additional tests have been conducted to evaluate the latency and robustness of the feature extractor under four undesirable operating scenarios.

Response to Reviewer 1

General Comment: This manuscript addresses an important and timely problem of autonomous physical interaction for tendon-driven aerial continuum manipulators using a hybrid vision/force control framework. However, The current version has substantial issues that prevent a confident assessment of the claimed fixed-time hybrid vision/force control performance. The main concerns are the strength of the theoretical assumptions, insufficient experimental comparison with baselines, incomplete reproducibility details, and several modeling inconsistencies in the vision-force coupling.

Response: The authors would like to thank the reviewer for this constructive and comprehensive comment. In response to the concerns raised, the manuscript has been substantially revised to strengthen the theoretical formulation, experimental validation, reproducibility, and consistency of the vision–force modeling. Specifically, the control architecture has been strengthened by incorporating a disturbance-rejection mechanism and reformulating the practical fixed-time hybrid vision/force control framework. The underlying theoretical assumptions have been carefully revisited and, where appropriate, either relaxed or more rigorously justified. In addition, the experiments have been repeated using the considered baseline methods under consistent conditions, enabling a more comprehensive and meaningful comparison of the proposed approach with existing methods. To improve reproducibility, several new tables have been included to provide detailed controller, system, and environmental parameters, together with additional implementation details. Finally, the modeling inconsistencies associated with the vision–force coupling have been systematically addressed, resulting in a unified and internally consistent formulation throughout the revised manuscript.

Comment 1.1: The proposed hybrid vision/force controller appears to offer limited conceptual novelty at the control-architecture level. Hybrid position/force or vision/force control has been extensively studied in aerial manipulation and robotic contact tasks. The manuscript's main distinction is the application of such a framework to a tendon-driven aerial continuum manipulator, together with line-based visual servoing and deep feature extraction. However, the controller itself is not sufficiently reformulated to address the distinctive coupled dynamics of this system.

Response:

We appreciate the reviewer's insightful comment and agree that hybrid position/force and vision/force control architectures have been extensively investigated for rigid manipulators and rigid-arm aerial manipulation systems. Accordingly, we have revised the manuscript to clarify that the novelty of the proposed approach does not reside in the general concept of combining vision and force feedback. Rather, it lies in the reformulation and integration of a robust hybrid interaction-control architecture specifically for a tendon-driven aerial continuum manipulator (TD-ACM), whose coupled kinematics, dynamics, actuation characteristics, compliance, and interaction behavior differ substantially from those of conventional rigid-arm aerial manipulators.

More specifically, the controller has been revised to explicitly account for the distinctive coupled dynamics and associated uncertainties of the TD-ACM. The adaptive super-twisting (ASTW) terms, $A_s(\sigma_s)$ and $A_f(\sigma_f)$, are introduced in the vision and force control channels, respectively, to compensate for lumped perturbations arising from model uncertainties, external disturbances, vehicle–manipulator dynamic coupling, tendon-driven actuation, and interaction effects. These terms are integrated with the proposed projection-based fixed-time sliding-mode formulation to establish practical fixed-time convergence of the vision and force tracking errors to prescribed residual sets.

In particular, the proposed controller is now explicitly written in terms of the strain-parameterized $SE(3)$ model of the coupled UAV-CR system. Unlike rigid-arm aerial manipulation, the tip Jacobian

is not obtained from a serial rigid-body chain, but from the strain-based propagation of the continuum backbone on $SE(3)$:

$$ {}^{I}\boldsymbol{g}_{a_i} = {}^{I}\boldsymbol{g}_{a_{i-1}} \exp\left(h_i \hat{\boldsymbol{\xi}}_{a_i}\right), \tag{1} $$

$$ {}^{a_i}\boldsymbol{J}_{a_i} = \mathrm{Ad}_{\left({}^{a_{i-1}}\boldsymbol{g}_{a_i}\right)^{-1}} \left(h_i \boldsymbol{T}_{a_i} \boldsymbol{B}_{a_\xi} \boldsymbol{\mathcal{E}}_a + {}^{a_{i-1}}\boldsymbol{J}_{a_{i-1}}\right), \tag{2} $$

As a result, the visual and force tasks are coupled to the deformable arm shape, the UAV motion, and the strain-dependent continuum kinematics.

The image-feature dynamics are then expressed as

$$ \dot{\boldsymbol{s}} = \hat{\boldsymbol{L}}\, \mathrm{Ad}_{c_{\boldsymbol{g}_t}}\, \boldsymbol{J}_t(\boldsymbol{q}_t)\dot{\boldsymbol{q}}_t + \boldsymbol{\Delta}_s, \tag{3} $$

or, equivalently,

$$ \dot{\boldsymbol{s}} = \boldsymbol{J}_L(\boldsymbol{q}_t, \boldsymbol{s})\dot{\boldsymbol{q}}_t + \boldsymbol{\Delta}_s, \qquad \boldsymbol{J}_L = \hat{\boldsymbol{L}}\, \mathrm{Ad}_{c_{\boldsymbol{g}_t}}\, \boldsymbol{J}_t. \tag{4} $$

Similarly, during contact, the normal-force rate is related to the same coupled TD-ACM motion through

$$ \dot{f}_n = \hat{k}_e \boldsymbol{Q}^{\,t} \boldsymbol{\varrho}_t + \Delta_f. \tag{5} $$

Using the visual differential kinematics, this gives the force-task Jacobian

$$ \dot{f}_n = \boldsymbol{J}_f(\boldsymbol{q}_t, \boldsymbol{s})\dot{\boldsymbol{q}}_t + \Delta_f, \tag{6} $$

where

$$ \boldsymbol{J}_f = \hat{k}_e \boldsymbol{Q} \left(\hat{\boldsymbol{L}}\, \mathrm{Ad}_{c_{\boldsymbol{g}_t}}\right)^{\dagger} \boldsymbol{J}_L. \tag{7} $$

Thus, the vision and force channels are not independent rigid-body tasks; both are mapped through the same continuum-arm Jacobian $\boldsymbol{J}_t$ and are affected by the same shape-dependent vehicle–arm coupling.

The visual and force error dynamics are written as

$$ \begin{aligned} \dot{\boldsymbol{e}}_s &= \dot{\boldsymbol{s}}_d - \boldsymbol{J}_L \dot{\boldsymbol{q}}_t - \boldsymbol{\Delta}_s, \\ \dot{e}_f &= \dot{f}_{dn} - \boldsymbol{J}_f \dot{\boldsymbol{q}}_t - \Delta_f. \end{aligned} \tag{8} $$

The proposed hybrid controller is reformulated as:

$$ \dot{\boldsymbol{q}}_{des} = \begin{cases} \boldsymbol{J}_L^{\dagger} \boldsymbol{v}_s, & \sigma_c = 0, \\ \dot{\boldsymbol{q}}_f + \boldsymbol{N}_f (\boldsymbol{J}_L \boldsymbol{N}_f)^{\dagger} \left(\boldsymbol{v}_s - \boldsymbol{J}_L \dot{\boldsymbol{q}}_f\right), & \sigma_c = 1, \end{cases} \tag{9} $$

where

$$ \dot{\boldsymbol{q}}_f = \boldsymbol{J}_f^{\dagger} \left(\dot{f}_{dn} + \Theta_f(e_f) + A_f(\sigma_f)\right), \tag{10} $$

$$ \boldsymbol{v}_s = \dot{\boldsymbol{s}}_d + \Theta_s(\boldsymbol{e}_s) + \boldsymbol{A}_s(\boldsymbol{\sigma}_s), \tag{11} $$

and

$$ \boldsymbol{N}_f = \boldsymbol{I} - \boldsymbol{J}_f^{\dagger} \boldsymbol{J}_f. \tag{12} $$

This formulation differs from a standard hybrid position/force controller because the force task is not imposed along a rigid manipulator direction. Instead, it is projected through the line-feature interaction matrix, the camera–tip adjoint transformation, and the continuum-arm Jacobian. Therefore, the controller directly accounts for the fact that the same deformable TD-ACM motion simultaneously changes the image features and the contact force.

Here, we clarify the connection between the outer-loop hybrid vision/force controller and the coupled TD-ACM dynamics. The full dynamics are given by

$$\boldsymbol{M}_t\ddot{\boldsymbol{q}}_t + (\boldsymbol{C}_t + \boldsymbol{D}_t)\dot{\boldsymbol{q}}_t + \boldsymbol{K}_t\boldsymbol{q}_t = \boldsymbol{B}_q\boldsymbol{\tau} + \boldsymbol{\mathcal{F}}_t, \tag{13}$$

where $\boldsymbol{M}_t$, $\boldsymbol{C}_t$, $\boldsymbol{D}_t$, and $\boldsymbol{K}_t$ are obtained from the strain-parameterized coupled model and include the UAV–continuum-arm inertial, damping, stiffness, and configuration-dependent coupling terms. Moreover, the actuation mapping is given by

$$\boldsymbol{B}_q = \begin{bmatrix} \boldsymbol{B}_{q_u} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{B}_{q_a} \end{bmatrix}, \qquad \boldsymbol{B}_{q_a} = \sum_{i_c=1}^{n_c} w_{i_c} \boldsymbol{B}_{a_\xi}^\top \boldsymbol{B}_{\tau_a}. \tag{14}$$

This term explicitly maps the tendon actuation into the strain coordinates of the continuum arm, which is absent in rigid-arm aerial manipulation.

To show how the high-level command is embedded into the coupled dynamics, let

$$\boldsymbol{e}_q = \boldsymbol{q}_t - \dot{\boldsymbol{q}}_{des}, \qquad \dot{\boldsymbol{q}}_r = \dot{\boldsymbol{q}}_{t,d} - \boldsymbol{\Lambda}_q\boldsymbol{e}_q, \tag{15}$$

and

$$\boldsymbol{S}_q = \dot{\boldsymbol{q}}_t - \dot{\boldsymbol{q}}_r = \dot{\boldsymbol{e}}_q + \boldsymbol{\Lambda}_q\boldsymbol{e}_q. \tag{16}$$

A low-level coupled dynamic tracking input may then be written as

$$\boldsymbol{B}_q\boldsymbol{\tau} = \hat{\boldsymbol{M}}_t\ddot{\boldsymbol{q}}_r + \left(\hat{\boldsymbol{C}}_t + \hat{\boldsymbol{D}}_t\right)\dot{\boldsymbol{q}}_r + \hat{\boldsymbol{K}}_t\boldsymbol{q}_t - \hat{\boldsymbol{\mathcal{F}}}_t - \boldsymbol{K}_S\boldsymbol{S}_q - \boldsymbol{K}_q\boldsymbol{e}_q + \hat{\boldsymbol{\Delta}}_q, \tag{17}$$

where $\hat{\boldsymbol{\Delta}}_q$ denotes the adaptive estimate of the lumped uncertainty in the low-level coupled dynamics. Substitution into the TD-ACM dynamics yields the closed-loop tracking-error equation

$$\boldsymbol{M}_t\dot{\boldsymbol{S}}_q + \boldsymbol{C}_t\boldsymbol{S}_q + \boldsymbol{K}_S\boldsymbol{S}_q + \boldsymbol{K}_q\boldsymbol{e}_q = \tilde{\boldsymbol{\Delta}}_q, \tag{18}$$

where $\tilde{\boldsymbol{\Delta}}_q$ collects the residual modeling and actuator uncertainties. This stability of this controller is well discussed in [1].

Changes in Manuscript:

- Reformulated the proposed controller in Theorem 2 and provided the corresponding stability proof in Appendix B.

Comment 1.2: A central concern is that the proposed method seems to treat the aerial platform motion, continuum-arm deformation, visual servoing loop, and contact-force regulation as largely separable components. This is problematic because the manuscript itself emphasizes that TD-ACMs exhibit strong vehicle–arm coupling, configuration-dependent compliance, deformation under contact, and arm dynamics occurring on time scales comparable to UAV attitude dynamics. These characteristics can affect the stability and accuracy of the entire closed-loop system, especially during dynamic motion and contact interaction. Under high-speed platform motion, aggressive attitude changes, or rapid contact transitions, the assumption that the visual and force loops can be designed through a reduced kinematic relationship may no longer hold. The coupling between UAV body motion, tendon-driven continuum deformation, and environmental contact could introduce unmodeled perturbations, delay-induced instability, or even loss of contact/control. The current stability proof does not appear to rigorously account for these coupled effects, actuator bandwidth limitations, UAV attitude-loop dynamics, or high-speed motion conditions.

Response:

We agree that the strong coupling among the aerial platform dynamics, continuum-arm deformation, visual servoing, and contact-force regulation must be explicitly considered when assessing the closed-loop behavior of a TD-ACM. To address this concern, the revised manuscript now includes the coupled dynamic equations of the TD-ACM, together with their derivation, and provides a clearer explanation of how the different components of the proposed control framework are integrated within the overall closed-loop system. We would also like to emphasize that the proposed framework is implemented as a unified architecture, as illustrated in Fig. 2, rather than as a collection of independent vision and force control loops. The force measurements obtained from the force/torque sensor are first processed using a robust adaptive Kalman filter (RAKF) to attenuate measurement noise and impulsive disturbances. In parallel, line features extracted from the eye-in-hand camera are processed by the singularity-aware line selection algorithm (SALS), which selects the most reliable line configuration while avoiding poorly conditioned or near-singular visual configurations.

More importantly, the revised dynamic formulation explicitly captures the coupling between the UAV platform and the tendon-driven continuum manipulator, including configuration-dependent effects and interaction forces. The residual effects that are not represented exactly by the nominal model—including modeling uncertainties, vehicle–arm coupling uncertainties, deformation effects, external disturbances, and contact-induced perturbations—are incorporated into the lumped perturbation terms of the closed-loop dynamics. The adaptive super-twisting (ASTW) terms in the vision and force channels are specifically introduced to compensate for these bounded perturbations. The corresponding stability analysis has been revised accordingly to establish practical fixed-time convergence in the presence of these uncertainties. This reduces the computational burden associated with constructing and processing a large interaction matrix while improving its numerical conditioning. This behavior is experimentally validated in Figs. 7 and 8, where the reciprocal condition number remains higher when SALS is employed compared with the case in which all detected lines are used without singularity-aware selection.

The coupled Jacobian of the TD-ACM is computed using the strain-based kinematic and dynamic models described in Sections II-B and II-C. The resulting image-feature error $\boldsymbol{e}_s$, normal-force tracking error e_f, and contact-mode signal σ_c are provided to the proposed ASTW–PFTSMC controller. The controller then generates the desired generalized velocity command $\dot{\boldsymbol{q}}_{\mathrm{des}}$.

At the low level, the desired generalized velocity is tracked using the adaptive controller proposed in [1]. Let $\dot{\boldsymbol{q}}_{des}$ denote the desired generalized configuration associated with the high-level velocity command, such that

$$\dot{\boldsymbol{q}}_{t,d} = \dot{\boldsymbol{q}}_{\mathrm{des}}. \tag{19}$$

The generalized-coordinate tracking error and reference velocity are defined as

$$\boldsymbol{e}_q = \boldsymbol{q}_t - \dot{\boldsymbol{q}}_{des}, \tag{20}$$

and

$$\dot{\boldsymbol{q}}_r = \dot{\boldsymbol{q}}_{t,d} - \boldsymbol{\Lambda}_q \boldsymbol{e}_q, \tag{21}$$

respectively, where $\boldsymbol{\Lambda}_q$ is a positive-definite diagonal matrix. Accordingly, the sliding variable is

$$\boldsymbol{S}_q = \dot{\boldsymbol{q}}_t - \dot{\boldsymbol{q}}_r = \dot{\boldsymbol{e}}_q + \boldsymbol{\Lambda}_q \boldsymbol{e}_q. \tag{22}$$

The low-level generalized control input is then expressed as

$$\boldsymbol{u}_t = \hat{\boldsymbol{M}}_t \ddot{\boldsymbol{q}}_r + \left(\hat{\boldsymbol{C}}_t + \hat{\boldsymbol{D}}_t\right) \dot{\boldsymbol{q}}_r + \hat{\boldsymbol{K}}_t \boldsymbol{q}_t - \boldsymbol{K}_v \dot{\boldsymbol{e}}_q - \boldsymbol{K}_p \boldsymbol{e}_q + \hat{\boldsymbol{\Delta}}_q, \tag{23}$$

where $\boldsymbol{u}_t = \boldsymbol{B}_q\boldsymbol{\tau}$, $\hat{\boldsymbol{M}}_t$, $\hat{\boldsymbol{C}}_t$, $\hat{\boldsymbol{D}}_t$, and $\hat{\boldsymbol{K}}_t$ denote the nominal estimates of the corresponding coupled dynamic matrices, $\boldsymbol{K}_p$ and $\boldsymbol{K}_v$ are positive-definite gain matrices, and $\hat{\boldsymbol{\Delta}}_q$ denotes the adaptive estimate of the lumped uncertainty associated with the low-level dynamics. The adaptation law is given by

$$\dot{\hat{\boldsymbol{\Delta}}}_q = -\boldsymbol{K}_{\Pi}\boldsymbol{S}_q, \tag{24}$$

where $\boldsymbol{K}_{\Pi}$ is a positive-definite adaptation-gain matrix. The control input is partitioned as

$$\boldsymbol{\tau} = \begin{bmatrix}\boldsymbol{\tau}_{uav}^{\top} & \boldsymbol{\tau}_{a}^{\top}\end{bmatrix}^{\top}, \tag{25}$$

where $\boldsymbol{\tau}_{uav} = [\tau_{uav_\phi}, \tau_{uav_\theta}, \tau_{uav_\psi}, \tau_{uav_x}, \tau_{uav_y}, \tau_{uav_z}]^{\top}$. The position-control component is converted into the desired total thrust and desired attitude commands as

$$f_z = \sqrt{\tau_{uav_x}^2 + \tau_{uav_y}^2 + \tau_{uav_z}^2}, \tag{26}$$

$$\theta_d = \tan^{-1}\left(\frac{\tau_{uav_x}\cos\psi_d + \tau_{uav_y}\sin\psi_d}{\tau_{uav_z}}\right), \tag{27}$$

and

$$\phi_d = \sin^{-1}\left(\frac{\tau_{uav_x}\sin\psi_d - \tau_{uav_y}\cos\psi_d}{\tau_{uav_z}}\right), \tag{28}$$

where ψ_d is the desired yaw angle. These desired attitude. The proposed high-level vision/force controller provides the desired generalized motion, while the low-level controller accounts for the underactuated nature of the aerial platform and converts the generalized command into feasible UAV and continuum-arm actuation commands.

We acknowledge the importance of robust low-level regulation under rapid platform motion and aggressive attitude variations. However, such operating conditions are less representative of the hybrid vision/force interaction tasks considered in this work, where physical interaction with relatively rigid objects generally requires slow or quasi-static platform motion to preserve stable contact. These more aggressive operating regimes have already been investigated in works such as [1, 2, 3] and are therefore not claimed as a contribution of the present study.

Nevertheless, unmodeled perturbations, sensing uncertainties, interaction-matrix uncertainty, force-model uncertainty, and delay-induced effects can directly degrade the outer-loop vision/force tracking performance. These effects are represented in the present formulation by the lumped uncertainty terms $\boldsymbol{\Delta}_s$ and Δ_f in the vision and force channels, respectively. The ASTW terms $\boldsymbol{A}_s(\boldsymbol{\sigma}_s)$ and $A_f(\sigma_f)$ are incorporated into the proposed high-level controller to reject these uncertainties, thereby enhancing the robustness of the hybrid vision/force interaction framework.

Changes in Manuscript:

- Added the coupled dynamic equations of the TD-ACM in Appendix A.
- Reformulated the proposed controller in Theorem 2 and provided the corresponding stability proof in Appendix B.

Comment 1.3: Theorem 1 claims boundedness of all closed-loop signals and fixed-time convergence of both visual and force errors. However, the proof in Appendix A relies heavily on Assumption 2, namely full row rank of the force-task Jacobian and the projected visual Jacobian throughout contact. This is a strong condition, but the manuscript does not provide practical verification that these rank conditions hold during simulation or experiments.

Response: We appreciate the reviewer's comment. First, we clarify that Theorem 1 has been revised to claim practical fixed-time stability, rather than exact fixed-time convergence, due to the residual errors associated with disturbance estimation. In addition, following the reviewer's recommendation, we have added the reciprocal condition number of the interaction matrix during the simulation to provide practical verification of the numerical conditioning of the visual servoing task.

Furthermore, a singularity-aware line selection algorithm (SALS) has been incorporated into the proposed framework. The purpose of SALS is to prevent the selected line configuration from becoming degenerate. Specifically, among the detected line features, SALS selects the subset that maximizes the reciprocal condition number of the interaction matrix. Therefore, instead of using all detected lines blindly, the controller uses a well-conditioned set of line features, which improves the robustness of the IBVS component near degenerate visual configurations.

Regarding kinematic singularities, one advantage of the proposed $SE(3)$-based strain-parameterized modeling framework is that the pose of each discretized continuum-arm section is propagated using the exponential map on $SE(3)$, rather than relying on local attitude coordinates such as Euler angles. Moreover, to avoid kinematic singularities corresponding to the straight CR configuration, the implementation uses the first-order small-angle approximation of the exponential map and its tangent-map derivative.

Changes in Manuscript:

- Added the Singularity-Aware Line Selector (SALS), presented in Algorithm 1.
- Added the following statement to Subsection II-B: "Moreover, to avoid numerical issues near the straight CR configuration, the first-order small-angle approximation of the $SE(3)$ exponential map is used."
- Conducted an additional experiment, namely Test #4: SALS Performance Assessment.

Comment 1.4: The interaction force model assumes $f_n = k_e p_t$, negligible friction, and normal interaction along the tip-frame z-axis. This is a severe simplification for aerial contact, especially because the paper itself notes aerodynamic disturbances and downwash effects. The assumption of negligible friction is stated by analogy with prior work, but no experimental evidence is provided to show that friction is indeed negligible for the presented setup. There is also a mathematical consistency issue. Equation (18) uses a pseudoinverse, whereas Eq. (22) and Eq. (28) use $(LAd_{(cg_t)})^{(-1)}$. This implies that the line-feature interaction matrix is square and nonsingular. The paper later states that only three nondegenerate line features are retained, giving six visual coordinates, but it does not define the nondegeneracy criterion or explain how singular or near-singular line configurations are detected and handled.

Response: We appreciate the reviewer's detailed comment. We agree that the original presentation of the force model and the line-feature singularity treatment was too compact. In the revised manuscript, we have clarified that the relation $f_n = \hat{k}_e p_t$ is not intended to represent a complete frictional contact model for aerial interaction. Rather, it is an approximate model used to regulate the measured normal component of the contact wrench. The same contact model is used in [4]. The tangential contact effects, frictional coupling, aerodynamic disturbances, downwash-induced perturbations, stiffness-estimation error, and force-sensor uncertainty are incorporated into the lumped uncertainty term of the force-channel dynamics.

Specifically, the coupled TD-ACM dynamics are written as

$$\boldsymbol{M}_t \ddot{\boldsymbol{q}}_t + (\boldsymbol{C}_t + \boldsymbol{D}_t)\dot{\boldsymbol{q}}_t + \boldsymbol{K}_t \boldsymbol{q}_t = \boldsymbol{B}_q \boldsymbol{\tau} + \boldsymbol{F}_t, \tag{29}$$

where

$$\boldsymbol{F}_t = \boldsymbol{S}_a^\top \boldsymbol{\mathcal{F}}_{p_a} + \boldsymbol{\mathcal{F}}_d + \boldsymbol{\mathcal{F}}_g. \tag{30}$$

Here, $\boldsymbol{\mathcal{F}}_{p_a}$ denotes the contact-point wrench exerted on the continuum-arm tip, while $\boldsymbol{\mathcal{F}}_d$ includes aerodynamic and other external disturbances. Therefore, aerodynamic and downwash-induced effects are not ignored in the dynamic formulation; they are included in the generalized disturbance wrench and then projected into the corresponding force/vision channels. As a result, during contact, the nominal normal-force model is written as

$$f_n = \hat{k}_e p_t + r_f, \qquad p_t \geq 0, \tag{31}$$

where $\hat{k}_e$ is the experimentally estimated environment stiffness and r_f represents the residual error caused by stiffness-estimation error, unmodeled compliance, frictional coupling, and measurement uncertainty. Taking the time derivative yields

$$\dot{f}_n = \hat{k}_e \boldsymbol{Q}^t \boldsymbol{\varrho}_t + \Delta_f, \tag{32}$$

where the lumped force-channel uncertainty can be expressed as

$$\Delta_f = (k_e - \hat{k}_e)\boldsymbol{Q}^t \boldsymbol{\varrho}_t + \dot{r}_f + \delta_{\text{fric}} + \delta_{\text{aero}} + \delta_{\text{sens}}. \tag{33}$$

Here, δ_{fric} represents the normal-channel effect induced by tangential/frictional contact components, δ_{aero} denotes the effect of aerodynamic/downwash disturbances, and δ_{sens} collects force-sensing and filtering residuals. Thus, the revised formulation does not require frictional and aerodynamic effects to be zero. Instead, it requires their projection into the controlled force channel to be bounded and continuously differentiable:

$$|\Delta_f| \leq \bar{\Delta}_f, \qquad |\dot{\Delta}_f| \leq \bar{\delta}_f. \tag{34}$$

We have also corrected the mathematical consistency issue pointed out by the reviewer. In the revised manuscript, all inverse operations associated with the line-feature interaction matrix and the force-task mapping are written using the Moore–Penrose generalized inverse. Specifically,

$$\dot{f}_n = \hat{k}_e \boldsymbol{Q} \left(\hat{\boldsymbol{L}} \, \mathrm{Ad}_{c_{\boldsymbol{g}_t}} \right)^\dagger \dot{\boldsymbol{s}} + \Delta_f, \tag{35}$$

and

$$\boldsymbol{J}_f = \hat{k}_e \boldsymbol{Q} \left(\hat{\boldsymbol{L}} \, \mathrm{Ad}_{c_{\boldsymbol{g}_t}} \right)^\dagger \boldsymbol{J}_L, \qquad \boldsymbol{J}_L = \hat{\boldsymbol{L}} \, \mathrm{Ad}_{c_{\boldsymbol{g}_t}} \boldsymbol{J}_t. \tag{36}$$

Therefore, the formulation no longer implies that $\hat{\boldsymbol{L}} \, \mathrm{Ad}_{c_{\boldsymbol{g}_t}}$ must be square and nonsingular.

To address the reviewer's concern regarding the retained three-line set, we have expanded the discussion of degenerate and near-degenerate line configurations [5]. For a candidate set of three lines, the selected interaction matrix is

$$\hat{\boldsymbol{L}}_{ijk} = \begin{bmatrix} \hat{\boldsymbol{L}}_i^\top & \hat{\boldsymbol{L}}_j^\top & \hat{\boldsymbol{L}}_k^\top \end{bmatrix}^\top \in \mathbb{R}^{6\times 6}. \tag{37}$$

Following the hidden-robot interpretation and the Grassmann–Cayley analysis of three observed image lines, the interaction matrix becomes singular when the corresponding wrench system loses rank. Let

$$\boldsymbol{\Xi} = \begin{bmatrix} \boldsymbol{\xi}_{11} & \boldsymbol{\xi}_{21} & \boldsymbol{\xi}_{31} & \boldsymbol{\xi}_{12} & \boldsymbol{\xi}_{22} & \boldsymbol{\xi}_{32} \end{bmatrix}, \tag{38}$$

where $\boldsymbol{\xi}_{i1} = [\boldsymbol{f}_{i1}^\top, \boldsymbol{0}^\top]^\top$ is the pure-force wrench associated with the ith line and $\boldsymbol{\xi}_{i2} = [\boldsymbol{0}^\top, \boldsymbol{m}_{i2}^\top]^\top$ is the corresponding pure-moment wrench. The general degeneracy conditions are

$$f_1 = \boldsymbol{f}_{11}^\top (\boldsymbol{f}_{21} \times \boldsymbol{f}_{31}) = 0, \tag{39}$$

or

$$f_2 = \boldsymbol{m}_{12}^{\top} (\boldsymbol{m}_{22} \times \boldsymbol{m}_{32}) = 0. \tag{40}$$

Changes in Manuscript:

- Eliminated the assumption of negligible friction and relaxed it to the boundedness of lumped disturbance forces.
- Revised Eq. (18) to use the Moore–Penrose pseudoinverse.
- Added the nondegeneracy conditions for line features in Definition 1, along with the corresponding reference.

Comment 1.5: Another concern is the use of plane parameters a_p, b_p, c_p, d_p. The manuscript states that the plane is expressed in the camera frame, but the simulation section assigns approximate constant plane parameters. If the camera moves relative to the target plane, these parameters should change unless an additional assumption or transformation is used.

Response: We appreciate the reviewer's careful observation. We agree that, in general, the plane parameters appearing in the line-feature interaction matrix are configuration dependent when expressed in the moving camera frame. In the revised manuscript, we have clarified that the constant plane parameters used in the simulation and experiments do not represent the exact instantaneous camera-frame plane parameters. Rather, they correspond to the desired camera configuration and are used to construct an estimated interaction matrix $\hat{\boldsymbol{L}}$, following the standard IBVS treatment in [6, 7].

Changes in Manuscript:

- Reformulated subsction III-A "Visual Servoing Based on Line Features" to use an estimate of the interaction matrix computed for the desired camera configuration, based on the similar approach by [6, 7].

Comment 1.6: The simulation comparison with PD-PI and CISMC is useful, but the experimental section does not provide comparable baseline experiments. The experiments demonstrate feasibility of the proposed method, but they do not establish that the proposed controller outperforms PD-PI or CISMC on the real hardware. The use of GlueStick is interesting and relevant, especially because line features are suitable for man-made environments. However, the paper presents GlueStick largely as a black-box deep feature extractor. There is no ablation against conventional line detectors, no analysis of feature-matching failure cases, and no quantitative assessment of feature-extraction latency. The manuscript gives useful hardware and simulation parameters, but many controller-critical parameters are missing. For example, the paper does not clearly report the values of C_1, C_2, M, N, J, Q, f_{th}, controller gains for PD-PI and CISMC, velocity limits, filtering settings, depth-estimation parameters, camera calibration details, and force-sensor filtering.

Response: We appreciate the reviewer's detailed comment. To address this concern, we repeated the hardware experiments for the PD–PI and CISMC controllers. The results of this hardware-level benchmarking are discussed in the revised experimental section. We also conducted an ablation study to assess the performance of GlueStick against conventional line detectors under various visually imperfect conditions, including illumination changes, blurring, and partial occlusion. In addition, the latency of each feature-extraction method is reported, and the factors contributing to feature-matching failure are discussed. The controller-critical parameters of the proposed method,

PD–PI, and CISMC, along with the camera calibration parameters and other relevant experimental settings, are listed in multiple tables in the revised manuscript.

Changes in Manuscript:

- Added Tables I and II, which list the plant and controller parameters.
- Conducted Test #3, "Line Detector Ablation Study."
- Conducted Test #5, "ASTW–PFTSMC Experimental Validation."

Response to Reviewer 2

Comment 2.1: Assumption 2 states that the force-task Jacobian maintains full row rank and the robot operates away from algorithmic singularities. While these are common assumptions, the paper would benefit from: Discussion of how close the system comes to singularities in practice, and what happens when the robot approaches singular configurations.

Response:

We appreciate the reviewer's comment. Overall, two sources of singularity may appear in the proposed framework: (i) image-space singularities associated with the selected line features and (ii) numerical singularities associated with the CR configuration. We have carefully revised the manuscript to address both issues.

Regarding the former, we rely on the analysis in [5], which reports the singularity loci associated with the observation of three image lines. These image-space singularities may occur in several geometric cases, including: (1) three coplanar lines with no common intersection point, where the plane containing the three lines contains the optical center or the optical center lies on the corresponding cylinder; (2) three coplanar lines with a common intersection point, for which the interaction matrix is never full rank; (3) three spatial lines with a common intersection point, which also leads to a rank-deficient interaction matrix; (4) three orthogonal spatial lines with no common intersection point, where the singularity loci are represented by quadric and cubic surfaces. These cases show that image-space singularities are mainly caused by the relative geometry between the observed line configuration and the camera.

To detect and avoid such configurations online, we have added the Singularity-Aware Line Selector (SALS). For each candidate triplet of line features, the corresponding interaction matrix is constructed as

$$\hat{\boldsymbol{L}}_{ijk} = \begin{bmatrix} \hat{\boldsymbol{L}}_i^\top & \hat{\boldsymbol{L}}_j^\top & \hat{\boldsymbol{L}}_k^\top \end{bmatrix}^\top, \tag{41}$$

and its reciprocal condition number is evaluated as

$$\mathcal{R}_{ijk} = \frac{\sigma_{\min}(\hat{\boldsymbol{L}}_{ijk})}{\sigma_{\max}(\hat{\boldsymbol{L}}_{ijk})}. \tag{42}$$

The triplet with the largest reciprocal condition number is selected for the controller. If this value falls below a prescribed threshold $\mathcal{R}_{\text{th}}$, or if fewer than three reliable line features are available, the controller activates the fallback strategy instead of applying the full IBVS law. Therefore, singular and near-singular image configurations are not assumed away; they are monitored through $\mathcal{R}_{ijk}$ and avoided through SALS.

The latter source of singularity is associated with numerical issues that may arise when the CR approaches a straight or zero-curvature configuration. In the strain-based formulation, the incremental motion of the ith continuum segment is computed from

$${}^{I}\boldsymbol{g}_{a_i} = {}^{I}\boldsymbol{g}_{a_{i-1}} \exp\left(h_i \hat{\boldsymbol{\xi}}_{a_i}\right), \tag{43}$$

where $\boldsymbol{\xi}_{a_i}$ is the strain vector at the ith discretization point and h_i is the spacing between two adjacent discretization points. Let

$$\theta_i = \|\boldsymbol{\Gamma}_{i,1:3}\| = h_i \|\boldsymbol{\xi}_{a_i,1:3}\|. \tag{44}$$

When $\theta_i \leq \theta_{\text{th}}$, where θ_{th} is a small positive scalar, the implementation uses the first-order small-angle approximation of the $SE(3)$ exponential map and its tangent-map derivative:

$$\boldsymbol{g}_i \approx \boldsymbol{I}_4 + \hat{\boldsymbol{\Gamma}}_i, \tag{45}$$

$$\boldsymbol{T}_{g_i} \approx \boldsymbol{I}_6 + \frac{1}{2}\,\mathrm{ad}_{\boldsymbol{\Gamma}_i}, \tag{46}$$

and

$$\dot{\boldsymbol{T}}_{g_i} \approx \frac{1}{2}\,\mathrm{ad}_{\dot{\boldsymbol{\Gamma}}_i}\,. \tag{47}$$

For $\theta_i > \theta_{\mathrm{th}}$, the closed-form $SE(3)$ exponential map and tangent-map expressions are used. This small-angle formulation prevents divisions by powers of θ_i when the rotation magnitude approaches zero and improves numerical robustness near straight CR configurations.

Accordingly, the revised manuscript addresses singularities at two levels: SALS handles image-space singularities caused by degenerate line configurations, while the small-angle $SE(3)$ implementation avoids numerical issues in the strain-based kinematics near zero curvature.

Changes in Manuscript:

- Added the Singularity-Aware Line Selector (SALS), presented in Algorithm 1.
- Added the following statement to Subsection II-B: "Moreover, to avoid numerical issues near the straight CR configuration, the first-order small-angle approximation of the $SE(3)$ exponential map is used."
- Conducted an additional experiment, namely Test #4: SALS Performance Assessment.

Comment 2.2: The contact force model assumes negligible friction (Assumption 1) and a simple spring-like behavior. While the authors cite prior work for justification, this simplification may be questionable for scenarios involving sliding contact (which appears relevant given the hybrid vision/force task), and non-planar surfaces or surfaces with varying stiffness. Please include a brief discussion of when this model might break down and how the robustness properties of the PFTSMC compensate for unmodeled friction effects.

Response: We appreciate the reviewer's detailed comment. We agree that the original presentation of the force model was not sufficiently accurate. In the revised manuscript, we have clarified that the relation $f_n = \hat{k}_e p_t$ is not intended to represent a complete frictional contact model for aerial interaction. Rather, it is an approximate control-oriented model used to regulate the measured normal component of the contact wrench. A similar normal-force model has also been used in [4].

Tangential contact effects, frictional coupling, aerodynamic disturbances, downwash-induced perturbations, stiffness-estimation error, and force-sensor uncertainty are incorporated into the lumped uncertainty term of the force-channel dynamics. Accordingly, the force model is written as

$$f_n = \hat{k}_e p_t + r_f, \qquad p_t \geq 0, \tag{48}$$

where r_f represents the residual modeling error caused by frictional coupling, non-planar contact, spatially varying stiffness, and sensing uncertainty. Therefore, the force-rate model used in the controller is expressed as

$$\dot{f}_n = \hat{k}_e \boldsymbol{Q}^{\,t} \boldsymbol{\varrho}_t + \Delta_f, \tag{49}$$

where Δ_f collects the unmodeled part of the force channel.

To compensate for this perturbation in the force channel, we have reformulated the controller by incorporating an adaptive super-twisting (ASTW) control law, which contributes to the rejection of the lumped disturbance. In addition, Assumption 1 has been revised to state that the unknown uncertainty should be bounded and continuously differentiable. Owing to these modifications, the revised controller preserves practical fixed-time stability in the presence of bounded force-channel perturbations.

Changes in Manuscript:

- Added the coupled dynamic equations of the TD-ACM in Appendix A.
- Reformulated the proposed controller in Theorem 2 and provided the corresponding stability proof in Appendix B.
- Revised Assumption 1.

Comment 2.3: How does the controller perform when the environment stiffness (k_e) is unknown or varies significantly? Is the fixed-time convergence guarantee still valid?

Response: Thank you for this important comment. We agree that the uncertainty or variation of the environment stiffness directly affects the validity of the nominal force model and, consequently, the achievable convergence property. In the revised manuscript, we have clarified that the controller does not require exact knowledge of k_e. Instead, the nominal stiffness $\hat{k}_e$ is used in the force-task Jacobian, while the mismatch between k_e and $\hat{k}_e$ is absorbed into the lumped force-channel perturbation Δ_f.

In the nominal formulation, the normal interaction force is modeled as

$$f_n = \hat{k}_e p_t + r_f, \qquad p_t \geq 0, \tag{50}$$

where p_t is the tip displacement along the controlled normal direction, $\hat{k}_e$ is the estimated environment stiffness, and r_f represents the residual modeling error. If the true stiffness is unknown or time varying, the actual force can be written more generally as

$$f_n = k_e(t) p_t + r_f. \tag{51}$$

Taking the time derivative gives

$$\dot{f}_n = k_e(t)\dot{p}_t + \dot{k}_e(t) p_t + \dot{r}_f. \tag{52}$$

Since the controlled normal velocity satisfies

$$\dot{p}_t = \boldsymbol{Q}^t \boldsymbol{\varrho}_t, \tag{53}$$

the force-rate dynamics can be rewritten as

$$\dot{f}_n = \hat{k}_e \boldsymbol{Q}^t \boldsymbol{\varrho}_t + \Delta_f, \tag{54}$$

where

$$\Delta_f = \left(k_e(t) - \hat{k}_e\right) \boldsymbol{Q}^t \boldsymbol{\varrho}_t + \dot{k}_e(t) p_t + \dot{r}_f + \delta_{\text{fric}} + \delta_{\text{aero}} + \delta_{\text{sens}}. \tag{55}$$

Thus, stiffness-estimation error, stiffness variation, unmodeled frictional effects, aerodynamic/downwash perturbations, and sensing residuals are all represented in the force-channel uncertainty term Δ_f.

Using the visual differential kinematics, the force dynamics are expressed as

$$\dot{f}_n = \boldsymbol{J}_f \dot{\boldsymbol{q}}_t + \Delta_f, \tag{56}$$

where the nominal force-task Jacobian is constructed using $\hat{k}_e$:

$$\boldsymbol{J}_f = \hat{k}_e \boldsymbol{Q} \left(\hat{\boldsymbol{L}} \operatorname{Ad}_{c\boldsymbol{g}_t}\right)^{\dagger} \boldsymbol{J}_L. \tag{57}$$

Accordingly, the force-error dynamics become

$$\dot{e}_f = \dot{f}_{dn} - \boldsymbol{J}_f \dot{\boldsymbol{q}}_t - \Delta_f. \tag{58}$$

The force component of the proposed controller is

$$\dot{\boldsymbol{q}}_f = \boldsymbol{J}_f^{\dagger}\left(\dot{f}_{dn} + \Theta_f(e_f) + A_f(\sigma_f)\right), \tag{59}$$

which yields the closed-loop uncertain force-error system

$$\dot{e}_f = -\Theta_f(e_f) - A_f(\sigma_f) - \Delta_f. \tag{60}$$

To compensate for this uncertainty, the force integral sliding variable is defined as

$$\sigma_f = e_f - e_f(0) + \int_0^t \Theta_f(e_f(\tau))\, d\tau. \tag{61}$$

Therefore,

$$\dot{\sigma}_f = -A_f(\sigma_f) - \Delta_f. \tag{62}$$

The adaptive super-twisting term is selected as

$$A_f(\sigma_f) = \alpha_f(t)\,\mathrm{sig}(\sigma_f)^{1/2} + \chi_f, \qquad \dot{\chi}_f = \frac{\beta_f(t)}{2}\,\mathrm{sign}(\sigma_f), \tag{63}$$

with

$$\beta_f(t) = 2\varepsilon_f \alpha_f(t). \tag{64}$$

Defining

$$\zeta_f = -\Delta_f - \chi_f, \tag{65}$$

one obtains

$$\begin{aligned} \dot{\sigma}_f &= -\alpha_f(t)\,\mathrm{sig}(\sigma_f)^{1/2} + \zeta_f, \\ \dot{\zeta}_f &= -\frac{\beta_f(t)}{2}\,\mathrm{sign}(\sigma_f) - \dot{\Delta}_f. \end{aligned} \tag{66}$$

Therefore, the convergence guarantee remains valid in the practical fixed-time sense provided that the stiffness variation and the resulting force-channel perturbation remain bounded, i.e.,

$$|\Delta_f| \le \bar{\Delta}_f, \qquad |\dot{\Delta}_f| \le \bar{\delta}_f. \tag{67}$$

This condition is satisfied when $k_e(t)$, $\dot{k}_e(t)$, p_t, ${}^t\boldsymbol{\varrho}_t$, and the unmodeled residual terms remain bounded during contact. Under this condition, the ASTW law compensates for the unknown stiffness-induced perturbation without requiring the exact value of k_e.

However, the revised theorem no longer claims exact fixed-time convergence in the presence of stiffness uncertainty. Instead, due to the residual perturbation estimation error, the force and visual errors are shown to be practically fixed-time stable. Specifically, after the ASTW transient, the force-error dynamics can be written as

$$\dot{e}_f = -\Theta_f(e_f) + r_f, \qquad |r_f| \le \eta_{f2}, \tag{68}$$

and, together with the visual channel,

$$\dot{\boldsymbol{e}}_s = -\Theta_s(\boldsymbol{e}_s) + \boldsymbol{r}_s, \qquad \|\boldsymbol{r}_s\| \le \bar{\eta}_s. \tag{69}$$

For the Lyapunov function

$$V = \frac{1}{2}\|\boldsymbol{e}_s\|^2 + \frac{1}{2}e_f^2, \tag{70}$$

the closed-loop system satisfies

$$\dot{V} \leq -\mathcal{L}_1 V^{\mathcal{B}_1} - \mathcal{L}_2 V^{\mathcal{B}_2} + \mathcal{D} V^{1/2}, \tag{71}$$

where $\mathcal{L}_1, \mathcal{L}_2, \mathcal{D} > 0$, $0 < \mathcal{B}_1 < 1$, and $\mathcal{B}_2 > 1$. Hence, the errors reach the compact residual set

$$\Omega = \left\{ (\boldsymbol{e}_s, e_f) : \mathcal{D} V^{1/2} \geq (1-\kappa) \left(\mathcal{L}_1 V^{\mathcal{B}_1} + \mathcal{L}_2 V^{\mathcal{B}_2} \right) \right\} \tag{72}$$

within the bounded time

$$T_p \leq \frac{1}{\kappa \mathcal{L}_1 (1-\mathcal{B}_1)} + \frac{1}{\kappa \mathcal{L}_2 (\mathcal{B}_2 - 1)}. \tag{73}$$

Therefore, partial knowledge of k_e does not invalidate the revised stability claim. It changes the result from exact fixed-time convergence to practical fixed-time convergence, with the residual bound depending on the magnitude of the stiffness-induced perturbation and the ASTW estimation residual. If the environment stiffness changes abruptly, becomes discontinuous, or varies beyond the boundedness and actuation assumptions, then the stated guarantee may no longer hold and a more detailed adaptive stiffness or contact-identification model would be required.

Changes in Manuscript:

- Added the coupled dynamic equations of the TD-ACM in Appendix A.
- Reformulated the proposed controller in Theorem 2 and provided the corresponding stability proof in Appendix B.
- Revised Assumption 1.

Comment 2.4: How does the proposed method handle line features that appear/disappear due to occlusions? The GlueStick framework uses matching confidence scores, but the control law assumes a fixed number of line features ($N_l = 3$). How is the transition handled when fewer than 3 lines are detected?

Response: Thank you for this important comment. We agree that the case in which the number of visible or reliably matched line features temporarily drops below three must be explicitly addressed. In the revised manuscript, we have clarified that $N_l = 3$ denotes the number of selected line features used by the proposed full-rank hybrid vision/force controller after the singularity-aware line selection algorithm (SALS), not the number of raw lines that must always be available in the image.

For a single line feature, the line interaction matrix satisfies

$$\hat{\boldsymbol{L}}_i \in \mathbb{R}^{2\times 6}, \tag{74}$$

and for N_l selected line features,

$$\hat{\boldsymbol{L}}_{N_l} = \begin{bmatrix} \hat{\boldsymbol{L}}_1^\top & \hat{\boldsymbol{L}}_2^\top & \cdots & \hat{\boldsymbol{L}}_{N_l}^\top \end{bmatrix}^\top \in \mathbb{R}^{2N_l \times 6}. \tag{75}$$

Therefore, when $N_l < 3$, one has

$$\operatorname{rank}(\hat{\boldsymbol{L}}_{N_l}) \leq 2N_l < 6, \tag{76}$$

which means that the visual task cannot fully constrain the six-dimensional camera twist. Thus, we do not claim that $\hat{\boldsymbol{L}}_{N_l}$ is full rank when fewer than three line features are available. The practical fixed-time stability result is stated for the intervals during which the selected visual interaction matrix satisfies the required rank and nondegeneracy conditions.

This solution is consistent with line-based IBVS aerial manipulation studies such as [7], where the authors explicitly note that the dimension of the visual interaction matrix changes with the number of detected lines. In that work, one-line and two-line cases are handled using reduced interaction matrices together with a complementary motion controller projected in the null-space of the visual task. In particular, for a reduced visual matrix $\hat{\boldsymbol{L}}$, the null-space projector is written as

$$\boldsymbol{P}_{vs} = \boldsymbol{I}_6 - \hat{\boldsymbol{L}}^{\dagger}\hat{\boldsymbol{L}}, \tag{77}$$

and the complementary command is introduced through this projector to constrain the visual-unobservable degrees of freedom. This confirms that when fewer than three line features are available, the missing degrees of freedom must be handled through an additional recovery or complementary control mechanism, which motivated us to add the SALS algorithm.

The proposed SALS module first filters the detected line set using the GlueStick matching confidence. For each candidate three-line subset, the corresponding 6×6 interaction matrix is constructed, and its reciprocal condition number is evaluated. The triplet with the largest reciprocal condition number is selected and accepted only when at least three reliable lines are available and $\mathcal{R}_\star \geq \mathcal{R}_{\text{th}}$. When fewer than three reliable lines are available or the selected triplet is near singular, the controller switches to a fallback recovery mode consisting of UAV hover, zero continuum-arm velocity, and a small yaw-search command to recover valid line observations. The SALS assessment shows improved conditioning compared with using all detected lines, with the reciprocal condition number increasing from approximately $\mathcal{R} = 0.055$ for the all-line strategy to $\mathcal{R} = 0.08$ with SALS. These results demonstrate that using more detected lines is beneficial only when they are selected based on reliability and conditioning.

Changes in Manuscript:

- Added the Singularity-Aware Line Selector (SALS), presented in Algorithm 1.
- Revised Remark 1 in subsection III-A to have discussion on cases when $N_l < 3$.

Comment 2.5: What is the physical interpretation of the strain parameterization used in the kinematics? The paper introduces this approach but does not provide intuition about how the strain variables relate to physical deformation.

Response: We appreciate the reviewer's comment. We agree that the physical meaning of the strain parametrization was not sufficiently explained in the original manuscript. In the revised manuscript, we have added a short explanation to clarify how the strain variables are related to the deformation of the continuum arm.

The strain vector is decomposed as

$$\boldsymbol{\xi}(X) = \begin{bmatrix} \boldsymbol{\xi}_a^{\top}(X) & \boldsymbol{\xi}_l^{\top}(X) \end{bmatrix}^{\top} \in \mathbb{R}^6, \tag{78}$$

where $\boldsymbol{\xi}_a(X) \in \mathbb{R}^3$ represents the angular strain and $\boldsymbol{\xi}_l(X) \in \mathbb{R}^3$ represents the linear strain. Physically, the components of $\boldsymbol{\xi}_a$ describe bending curvatures about the local transverse axes and torsion about the local backbone direction. The components of $\boldsymbol{\xi}_l$ describe axial stretch along the backbone and transverse shear deformations. Therefore, the strain variables directly encode the local bending, twisting, stretching, and shearing deformation of the continuum arm.

Changes in Manuscript:

- Added a physical interpretation of the strain parametrization in Subsection II-B.

Comment 2.6: How sensitive is the controller to errors in the estimated depth information? The IBVS method requires depth information which is typically difficult to estimate accurately in monocular systems.

Response: The authors acknowledge the reviewer's concern regarding the sensitivity of IBVS to depth-estimation errors, particularly in monocular vision systems. In the proposed formulation, the plane parameters associated with the observed line features are expressed in the camera frame and therefore vary with the camera pose. Consequently, their exact values cannot, in general, be determined accurately throughout the maneuver. To address this issue, we adopt an uncertain image interaction matrix, following the approach suggested by Espiau et al. [6] and subsequently applied in [7]. Specifically, the interaction matrix is evaluated using the plane parameters corresponding to the desired camera configuration and is kept fixed during execution. As the camera moves away from the desired configuration, the discrepancy between the nominal and actual plane/depth parameters manifests itself as uncertainty in the interaction matrix. This formulation therefore avoids requiring accurate online depth estimation from the monocular camera. In the revised manuscript, this modeling uncertainty is explicitly accounted for within the robust control formulation. The resulting interaction-matrix uncertainty is treated as a bounded perturbation, whose effect is compensated by the robust terms of the proposed controller. This approach reduces the controller's dependence on precise depth information while preserving the stated practical fixed-time convergence properties under the assumed bounded uncertainty conditions.

Changes in Manuscript:

- Reformulated subsction III-A "Visual Servoing Based on Line Features" to use an estimate of the interaction matrix computed for the desired camera configuration, based on the similar approach by [6, 7].

Response to Reviewer 3

Comment 3.1: In the contribution, the authors state the proposed method "enhances robustness to viewpoint variations, partial occlusions, and illumination changes". The authors should provide a theoretical characterization of how these disturbances propagate into the control loop, and quantify the control performance degradation under these adverse conditions.

Response: The authors acknowledge the reviewer's concern regarding the sensitivity of IBVS to depth-estimation errors, particularly in monocular vision systems. In the proposed formulation, the plane parameters associated with the observed line features are expressed in the camera frame and therefore vary with the camera pose. Consequently, their exact values cannot, in general, be determined accurately throughout the maneuver. To address this issue, we adopt an uncertain image interaction matrix, following the approach suggested by Espiau et al. [6] and subsequently applied in [7]. Specifically, the interaction matrix is evaluated using the plane parameters corresponding to the desired camera configuration and is kept fixed during execution. As the camera moves away from the desired configuration, the discrepancy between the nominal and actual plane/depth parameters manifests itself as uncertainty in the interaction matrix. This formulation therefore avoids requiring accurate online depth estimation from the monocular camera. In the revised manuscript, this modeling uncertainty is explicitly accounted for within the robust control formulation. The resulting interaction-matrix uncertainty is treated as a bounded perturbation, whose effect is compensated by the robust terms of the proposed controller. This approach reduces the controller's dependence on precise depth information while preserving the stated practical fixed-time convergence properties under the assumed bounded uncertainty conditions.

Changes in Manuscript:

- Reformulated the image-feature dynamics in Section III to include the lumped visual uncertainty term $\boldsymbol{\Delta}_s$.
- Reformulated the proposed ASTW–PFTSMC controller in Theorem 2 and provided the corresponding practical fixed-time stability proof in Appendix B.
- Conducted Test #3, "Line Detector Ablation Study," to quantify feature-matching robustness under nominal, severe illumination, blurring, and partial-occlusion conditions.
- Added the feature-detection and matching latency comparison among the investigated line detectors.

Comment 3.2: A hybrid vision/force controller is proposed as an innovative design. I suggest to further clarify the coupling mechanism between the vision and force control loops, what causes this coupling, and how it introduces key challenges for the hybrid controller design.

Response: We acknowledge the reviewer's comment and have clarified this point in the revised Introduction. Specifically, we added a discussion explaining that, in contact-rich IBVS tasks, the visual and force loops are intrinsically coupled because the same tip/camera motion simultaneously changes the observed image features and the normal contact force. As a result, a command generated to reduce the visual error may disturb force regulation, while a command generated to regulate the normal force may alter the image features. This creates a conflicting control objective that must be explicitly handled in the hybrid controller design.

To address this challenge, we further clarified that the proposed controller resolves the coupling through a force-employs a projection-based fixed-time sliding mode controller (PFTSMC). During contact, the normal-force regulation is assigned to the force task, while the visual-servoing command is projected into the null space of the force task. This prevents the visual controller from directly

disturbing the regulated normal-force direction while preserving visual correction in the admissible subspace. The revised manuscript also revised the controller formulation to include uncertainty in the image interaction matrix, uncertainty in the contact stiffness and force model, contact switching, and possible rank deficiency or singularity of the projected visual Jacobian. These challenges motivate the use of the SALS module and the ASTW disturbance rejection terms in the proposed framework.

Changes in Manuscript:

- Added a discussion in the Introduction clarifying the intrinsic coupling between the visual-servoing and force-regulation loops in contact-rich IBVS tasks.
- Revised Fig. 2 to illustrate the integration of image feedback, force feedback, RAKF, CDA, SALS, differential kinematics, and the ASTW–PFTSMC controller.
- Reformulated the hybrid vision/force controller in Theorem 2 using the force-task Jacobian $\boldsymbol{J}_f$ and the null-space projector $\boldsymbol{N}_f$.

Comment 3.3: The authors set the parameter N_l as $N_l = 3$, which may be a little simplified. The authors should analyze system stability when N_l temporarily drops below 3 due to occlusion or feature matching. If the feature is less than 2, how to ensure matrix L is full rank. The authors should also analyze how the proposed method performs with more line features, and test the performance with experiments.

Response: Thank you for this important comment. We agree that the case in which the number of visible or reliably matched line features temporarily drops below three must be explicitly addressed. In the revised manuscript, we have clarified that $N_l = 3$ denotes the number of selected line features used by the proposed full-rank hybrid vision/force controller after the singularity-aware line selection algorithm (SALS), not the number of raw lines that must always be available in the image.

For a single line feature, the line interaction matrix satisfies

$$\hat{\boldsymbol{L}}_i \in \mathbb{R}^{2\times 6}, \tag{79}$$

and for N_l selected line features,

$$\hat{\boldsymbol{L}}_{N_l} = \begin{bmatrix} \hat{\boldsymbol{L}}_1^\top & \hat{\boldsymbol{L}}_2^\top & \cdots & \hat{\boldsymbol{L}}_{N_l}^\top \end{bmatrix}^\top \in \mathbb{R}^{2N_l\times 6}. \tag{80}$$

Therefore, when $N_l < 3$, one has

$$\operatorname{rank}(\hat{\boldsymbol{L}}_{N_l}) \le 2N_l < 6, \tag{81}$$

which means that the visual task cannot fully constrain the six-dimensional camera twist. Thus, we do not claim that $\hat{\boldsymbol{L}}_{N_l}$ is full rank when fewer than three line features are available. The practical fixed-time stability result is stated for the intervals during which the selected visual interaction matrix satisfies the required rank and nondegeneracy conditions.

This treatment is consistent with line-based IBVS aerial manipulation studies such as [7], where the authors explicitly note that the dimension of the visual interaction matrix changes with the number of detected lines. In that work, one-line and two-line cases are handled using reduced interaction matrices together with a complementary motion controller projected in the null-space of the visual task. In particular, for a reduced visual matrix $\hat{\boldsymbol{L}}$, the null-space projector is written as

$$\boldsymbol{P}_{vs} = \boldsymbol{I}_6 - \hat{\boldsymbol{L}}^\dagger \hat{\boldsymbol{L}}, \tag{82}$$

and the complementary command is introduced through this projector to constrain the visual-unobservable degrees of freedom. This confirms that when fewer than three line features are available, the missing degrees of freedom must be handled through an additional recovery or complementary control mechanism, which motivated us to add the SALS algorithm.

The proposed SALS module first filters the detected line set using the GlueStick matching confidence. For each candidate three-line subset, the corresponding 6×6 interaction matrix is constructed, and its reciprocal condition number is evaluated. The triplet with the largest reciprocal condition number is selected and accepted only when at least three reliable lines are available and $\mathcal{R}_\star \geq \mathcal{R}_{\text{th}}$. When fewer than three reliable lines are available or the selected triplet is near singular, the controller switches to a fallback recovery mode consisting of UAV hover, zero continuum-arm velocity, and a small yaw-search command to recover valid line observations. Finally, we added experimental evidence to support the proposed strategy. The SALS assessment shows improved conditioning compared with using all detected lines, with the reciprocal condition number increasing from approximately $\mathcal{R} = 0.055$ for the all-line strategy to $\mathcal{R} = 0.08$ with SALS. These results demonstrate that using more detected lines is beneficial only when they are selected based on reliability and conditioning.

Changes in Manuscript:

- Revised Remark 1 in Subsection III-A to clarify that $N_l = 3$ denotes the selected line set and to discuss the case when $N_l < 3$.
- Added the nondegeneracy conditions for three line features in Definition 1.
- Added the Singularity-Aware Line Selector (SALS), presented in Algorithm 1.
- Conducted Test #4, "SALS Performance Assessment," to compare the reciprocal condition number obtained by SALS with the all-line strategy.

Comment 3.4: The switching logic in (23) is binary and threshold-based. However, the behavior during the transition from free motion to contact is not analyzed. Is there any chattering and how to prevent it? How is the threshold f_{th} selected?

Response: Thank you for this important comment. We agree that a binary switching law may lead to chattering if it is implemented using a single force threshold. In the revised manuscript, we have clarified that the contact mode is not switched directly from the instantaneous force measurement. Instead, the switching variable σ_c is generated by the contact detection algorithm (CDA), which uses the filtered normal force, a hysteresis band, and a consecutive-frame validation rule. Here, we have also clarified how the two thresholds, namely the contact activation and contact release thresholds f_L, are selected. It is worth noting that the original manuscript employed only a single threshold f_{th}, whereas the revised algorithm incorporates two distinct thresholds.

Specifically, the measured force is first processed by the robust adaptive Kalman filter (RAKF), and the filtered normal force is denoted by $f_n(k)$ at the kth sampling instant. The contact mode is then determined by two thresholds,

$$f_L < f_{th}, \tag{83}$$

where f_{th} is the contact-activation threshold and f_L is the contact-release threshold. The counters N_{on} and N_{off} are updated as

$$\begin{aligned} f_n(k) \geq f_{th} &\Rightarrow N_{\text{on}}(k) = N_{\text{on}}(k-1) + 1, \quad N_{\text{off}}(k) = 0, \\ f_n(k) \leq f_L &\Rightarrow N_{\text{off}}(k) = N_{\text{off}}(k-1) + 1, \quad N_{\text{on}}(k) = 0. \end{aligned} \tag{84}$$

The contact mode is updated only after the corresponding condition holds for N_h consecutive samples:

$$\sigma_c(k) = \begin{cases} 1, & N_{\text{on}}(k) \geq N_h, \\ 0, & N_{\text{off}}(k) \geq N_h, \\ \sigma_c(k-1), & \text{otherwise.} \end{cases} \tag{85}$$

Therefore, short force spikes, sensor noise, and transient contact events do not immediately change the control mode. The hysteresis condition $f_L < f_{th}$ prevents repeated switching near the contact boundary, while the consecutive-frame condition prevents chattering caused by isolated noisy force samples.

Using this contact-state variable, the hybrid command is applied as

$$\dot{\boldsymbol{q}}_{des} = \begin{cases} \boldsymbol{J}_L^{\dagger} \boldsymbol{v}_s, & \sigma_c = 0, \\ \dot{\boldsymbol{q}}_f + \boldsymbol{N}_f (\boldsymbol{J}_L \boldsymbol{N}_f)^{\dagger} \left(\boldsymbol{v}_s - \boldsymbol{J}_L \dot{\boldsymbol{q}}_f \right), & \sigma_c = 1, \end{cases} \tag{86}$$

where the force-control command is activated only after the CDA confirms sustained contact. Equivalently, the transition can be viewed as

$$\dot{\boldsymbol{q}}_{des} = (1 - \sigma_c) \boldsymbol{J}_L^{\dagger} \boldsymbol{v}_s + \sigma_c \left[\dot{\boldsymbol{q}}_f + \boldsymbol{N}_f (\boldsymbol{J}_L \boldsymbol{N}_f)^{\dagger} \left(\boldsymbol{v}_s - \boldsymbol{J}_L \dot{\boldsymbol{q}}_f \right) \right]. \tag{87}$$

Thus, the switching is binary at the controller level, but the decision signal σ_c is filtered through the CDA logic rather than being generated by a single instantaneous threshold comparison.

The threshold f_{th} is selected from the no-contact force-noise level and the minimum force that reliably indicates physical contact. In practice, the filtered no-contact force is recorded before the experiment, and its mean and standard deviation are denoted by $\mu_{f,0}$ and $\sigma_{f,0}$, respectively. The activation threshold is chosen as

$$f_{th} = \mu_{f,0} + \kappa_{\text{on}} \sigma_{f,0}, \tag{88}$$

with $\kappa_{\text{on}} > 1$, so that normal sensor noise and small aerodynamic fluctuations do not activate the contact mode. The release threshold is selected lower than the activation threshold,

$$f_L = \mu_{f,0} + \kappa_{\text{off}} \sigma_{f,0}, \qquad \kappa_{\text{off}} < \kappa_{\text{on}}, \tag{89}$$

which creates the hysteresis band

$$\mathcal{H}_f = [f_L,\ f_{th}]. \tag{90}$$

In the experiments, the values

$$f_L = 0.023 \text{ N}, \qquad f_{th} = 0.0381 \text{ N}, \qquad N_h = 5 \tag{91}$$

were used. These values were selected to be above the filtered force noise floor while remaining small compared with the desired contact-force level, so that contact is detected early but not triggered by isolated measurement spikes.

Contact to free-flight transition stability: Let $T_f > 0$ denote the sampling period of the force measurement and $\{t_j\}_{j=1}^{\infty}$ be the sequence of switching instants generated by the contact detection algorithm (CDA). As σ_c is changed only at sampling instants and each transition needs $\mathcal{N}_h$ consecutive samples meeting the related threshold condition, the time between two opposite transitions is limited as follows:

$$t_{j+1} - t_j \geq \tau_d, \qquad \tau_d = \mathcal{N}_h T_f. \tag{92}$$

Therefore, infinitely many switches cannot occur in a finite time interval, and the CDA excludes Zeno behavior. Moreover, because $f_{\mathrm{L}} < f_{\mathrm{th}}$, the interval $[f_{\mathrm{L}}, f_{\mathrm{th}}]$ forms a hysteresis band. Therefore, a single noisy force sample or a force value inside this band cannot change the contact mode.

At a switching instant t_j, only the discrete variable σ_c changes. The physical TD-ACM states, the image features, and the measured normal force remain continuous. Hence,

$$\boldsymbol{e}_s(t_j^+) = \boldsymbol{e}_s(t_j^-), \qquad e_f(t_j^+) = e_f(t_j^-). \tag{93}$$

During the free-motion mode ($\sigma_c = 0$), the active Lyapunov function is

$$V_0 = \frac{1}{2}\|\boldsymbol{e}_s\|^2. \tag{94}$$

During the contact mode ($\sigma_c = 1$), the active Lyapunov function is

$$V_1 = \frac{1}{2}\|\boldsymbol{e}_s\|^2 + \frac{1}{2}e_f^2. \tag{95}$$

For a contact-to-free transition $\sigma_c : 1 \to 0$, one obtains

$$V_0(t_j^+) = \frac{1}{2}\|\boldsymbol{e}_s(t_j)\|^2 \le \frac{1}{2}\|\boldsymbol{e}_s(t_j)\|^2 + \frac{1}{2}e_f^2(t_j) = V_1(t_j^-). \tag{96}$$

Therefore, release from contact cannot increase the active Lyapunov function.

For a free-to-contact transition $\sigma_c : 0 \to 1$, the force channel is activated. Since the desired force and the filtered measured force are bounded, there exists a finite constant $\bar{e}_f > 0$ such that

$$|e_f(t_j)| \le \bar{e}_f. \tag{97}$$

Thus,

$$V_1(t_j^+) = V_0(t_j^-) + \frac{1}{2}e_f^2(t_j) \le V_0(t_j^-) + \frac{1}{2}\bar{e}_f^2. \tag{98}$$

Hence, contact engagement may introduce only a bounded change in the active Lyapunov function, not an unbounded or impulsive growth.

Changes in Manuscript:

- Added the Contact Detection Algorithm (CDA), presented in Algorithm 2.
- Added the CDA parameters ($f_{\mathrm{L}}, f_{\mathrm{th}}, \mathcal{N}_h$) to Table II.
- Added Test #2, "Force Filtering and CDA Performance Evaluation," to show the filtered force signal and the detected contact interval.
- Revised the controller description to clarify that the switching variable σ_c is generated by the CDA using RAKF-filtered force, hysteresis, and a consecutive-sample rule.

Comment 3.5: Fig. 1 can be improved to be more informative. I suggest to illustrate some key variables in the figure, such as θ, ρ, and others.

Response: The suggested variables are added to Fig. 1 to allow for more clarity.

Changes in Manuscript:

- Revised Fig. 1 to include additional key variables, including the line-feature parameters θ_i and ρ_i, and the relevant coordinate-frame information.

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